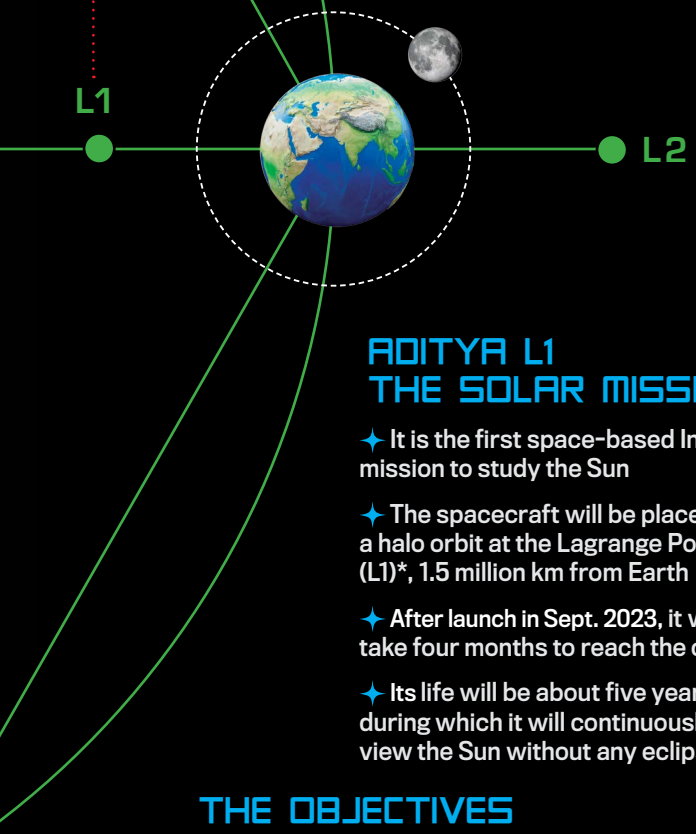




ADITYA L1 THE SOLAR MISSION

- It is the first space-based Indian mission to study the Sun
- The spacecraft will be placed in a halo orbit at the Lagrange Point 1 (L1)*, 1.5 million km from Earth
- After launch in Sept. 2023, it will take four months to reach the orbit
- Its life will be about five years, during which it will continuously view the Sun without any eclipses

THE OBJECTIVES

- The spacecraft will carry seven payloads to observe the Sun's photosphere, chromosphere and corona, using electromagnetic, particle and magnetic field detectors
- Will mainly study the problem of coronal heating, coronal mass ejection, pre-flare and flare activities and their characteristics

**Lagrange points (L1-L5) are positions in space where the gravitational pull of the Sun and Earth equals the centripetal force required for a spacecraft to move in tandem, allowing it to observe the Sun continuously without consuming too much fuel*

Having conquered the moon, the Indian Space Research Organisation (ISRO) is now trying to find its place under the sun. Even as the Chandrayaan-3 lander and rover were busy carrying out their scientific tasks on the lunar surface, other ISRO scientists and engineers were focusing on preparing the Aditya L1 spacecraft for its long journey to study the Sun. India's first mission to explore our nearest star works on the same principles as the Chandrayaan mission, where ISRO deployed innovative methods and ensured frugality in financial resources. America's National Aeronautics and Space Administration (NASA) and the European Space Agency (ESA) have already carried out over 20 scientific space missions to the Sun, but ISRO, as it did with Chandrayaan, wants to come up with its own unique discoveries. Especially on the composition of the Sun's corona, photosphere and chromosphere. The space organisation hopes to do so by using the scientific instruments aboard Aditya to measure the solar electromagnetic fields and particle ejections that could shed new light on its behaviour.

The other major interplanetary project on ISRO's radar is sending an orbiter to Venus to study the planet's composition and atmosphere. Again, only a handful of countries have sent space probes to the planet nearest to us after Mercury.

Graphic by TANMOY CHAKRABORTY